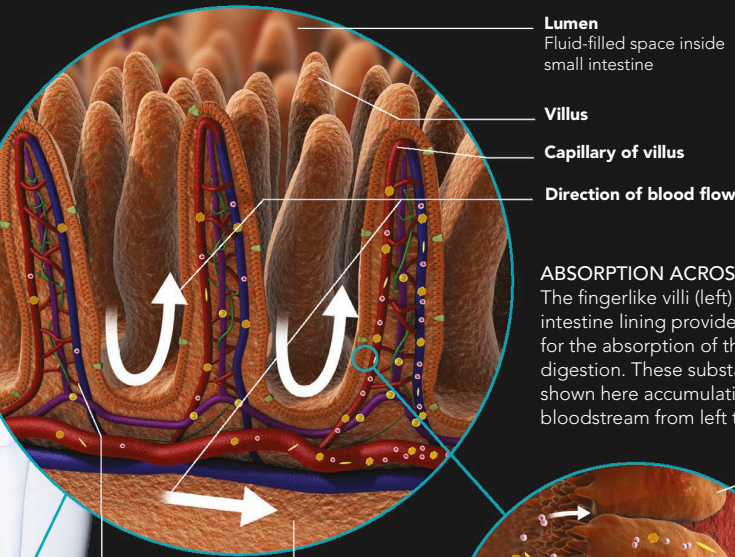




ABSORPTION ACROSS VILLI

The fingerlike villi (left) of the small intestine lining provide a large area for the absorption of the products of digestion. These substances are shown here accumulating in the bloodstream from left to right.

Lacteal
Lymph
capillary
of villus

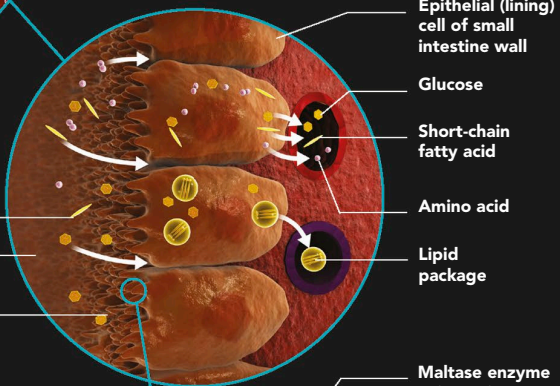
**Wall of
small
intestine**

Fatty acid
Small intestine lumen

Epithelial cell membrane
Formed into "brush" of microvilli

CLOSE-UP OF VILLUS SURFACE

Short-chain fatty acids, glucose, and amino acids pass through the epithelial cells into a capillary (red). Larger fatty acids are packaged and passed into a lymph lacteal (purple).



Maltase enzyme
Splits maltose
into glucose

Glucose
Passes through
protein channel

**Peptidase
enzyme**
Splits peptides
into amino acids

Amino acids
Pass through
protein channel in
twos and threes

**Epithelial cell
interior**

EXTREME CLOSE-UP OF CELL MEMBRANE

Enzymes that complete digestion are embedded in the membrane of the epithelial cells (right). The resulting sugars and amino acids pass through dedicated protein channels, while fatty acids diffuse across.

**Short-chain
fatty acid**
Simply diffuses
across cell
membrane

**Small intestine
lumen**

**Epithelial cell
membrane**

